

EXPERIMENT 3

Practical 3: Traversing and Map Preparation

Aim: To carry out a closed traverse with a theodolite, and prepare a map of the area along with a contour map using the concepts of Gale's Traverse Table.

Apparatus Required:

- Transit Theodolite with tripod
- Measuring tape (30m) or chain
- Ranging rods
- Leveling staff
- Plumb bob
- Pegs and arrows

Theory:

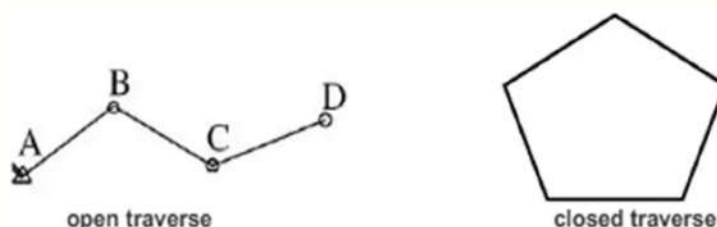
A traverse consists of a series of connected survey lines whose lengths and directions are measured. There are two primary types of traverses:

- **Open Traverse:** A traverse that starts at a known point but ends at a different, unknown point. Because it does not loop back on itself, it is difficult to mathematically check for field errors. It is generally used for long, linear projects like roads, railways, or pipelines.
- **Closed Traverse:** A traverse that starts and ends at the exact same starting point, forming a closed polygon. Because the start and end points are identical, this method allows for rigorous mathematical checks to identify and correct any surveying errors.

Gale's Traverse Table is a systematic method used to compute and balance the coordinates of the stations in a closed traverse. It involves calculating the consecutive coordinates (Latitudes and Departures) of the traverse legs.

- **Latitude (L):** $L \cos(\theta)$
- **Departure (D):** $L \sin(\theta)$ (Where L is the length of the line and θ is the Reduced Bearing)

Closing errors are distributed using balancing rules, most commonly Bowditch's Rule (Compass Rule) or the Transit Rule, before plotting the final independent coordinates.



Procedure:

1. **Reconnaissance:** Inspect the area and establish a closed circuit of stations (e.g., A, B, C, D, E) using wooden pegs. Ensure intervisibility between adjacent stations.
2. **Linear Measurements:** Accurately measure the lengths of all traverse legs (AB, BC, CD, etc.) using a steel tape.

- ### Observation Table:

GALE'S TRAVERSE TABLE																			
Station	Line Length (m)	Angles	Correction	Corrected Angles	WCB	Consecutive Coordinates				Corrections				Corrected Consecutive Coordinates				Independent Coordinates	
						N	S	E	W	N	S	E	W	N	S	E	W	N	E
A	24.75 (AB)	69°36'40"	0°2'19"	69°34'21"	358°40'	24.74			0.58	+0.32		+0.05		25.06			0.63	300	500
B	15.90 (BC)	124°30'30"	0°2'19"	124°28'11"	51°1'49"	9.30		12.90		+0.21			-0.03	9.51		12.86		325.06	499.37
C	22.25 (CD)	123°25'10"	0°2'19"	123°22'51"	110°56'58"		7.91	20.80			-0.29	-0.04			7.62	20.76		334.57	512.23
D	15.90 (DE)	100°07'25"	0°2'19"	100°5'6"	190°43'52"		15.62		2.96	-0.21			+0.03		15.41		2.99	326.95	532.99
E	32.25 (EA)	122°31'50"	0°2'19"	122°29'31"	248°14'21"	11.95		29.95		-0.42		+0.06		11.54			30.01	311.54	530.00
																		300	500
Perimeter of Traverse		111.06				34.64	35.49	33.70	33.49										
Sum of all Included angles		540°11'35"		540°		ΔL = EN - ES		ΔD = EE - EW											
Total Correction		0°11'35"				ΔL = -1.45		ΔD = +0.21											
Correction in each angle		0°2'19"																	

Precautions:

1. The theodolite must be accurately centered over the station peg and perfectly leveled before taking any angle.
2. Always eliminate parallax by properly focusing the eyepiece and the objective lens.
3. Both verniers (A and B) should be read to eliminate any errors due to the eccentricity of the instrument's centers.
4. Always rotate the theodolite in a clockwise direction to avoid backlash errors in the tangent screws.

Conclusion: The closed traverse was successfully completed and balanced using Bowditch's rule, the calculated closing error of 0.553 m falls well within the permissible limits for standard surveying.

*** Abbreviations-

- **WCB:** Whole Circle Bearing (The angle measured clockwise from North, ranging from 0° to 360°)
- **RB:** Reduced Bearing (The angle measured from either North or South towards East or West, ranging from 0° to 90°)
- **Lat:** Latitude (The North-South component of a survey line)
- **Dep:** Departure (The East-West component of a survey line)
- **Corr:** Correction (The mathematical adjustment applied to fix closing errors)
- **Bal:** Balanced (The final, corrected Latitude or Departure after errors are distributed)
- **Indep:** Independent Coordinates (The absolute coordinates of a station measured from a single common origin point)
- **N, S, E, W:** North, South, East, West (Used for directions and quadrants)
- **ΔL :** Error in Latitude (The total closing error on the North-South axis)
- **ΔD :** Error in Departure (The total closing error on the East-West axis)
- **EN** = Sum of North Latitudes
- **ES** = Sum of South Latitudes
- **ED** = Error in Departure
- **EE** = Sum of East Departures
- **EW** = Sum of West Departures
- **Δ :** Total Closing Error (The final combined error before correction)

EXPERIMENT 6

Practical 6: Total Station and GPS

Aim: To carry out an open traverse using a Total Station and Global Positioning System (GPS).

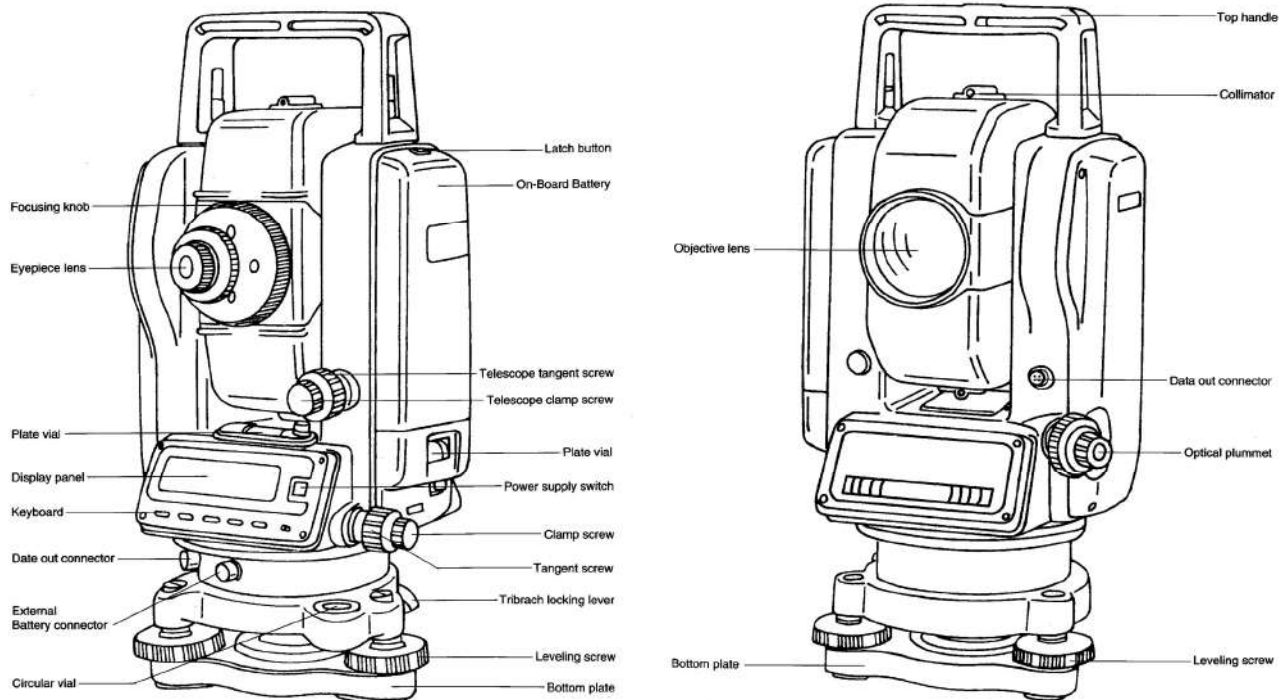
Apparatus Required:

- Total Station with tripod
- Prism reflector with prism pole
- Handheld GPS receiver or DGPS setup
- Wooden pegs and marker pen

Theory:

An open traverse consists of a series of connected lines that do not return to the starting point. It is typically used for surveying route projects like roads, railways, or pipelines.

- **Total Station:** A Total Station is an advanced electronic surveying instrument that combines a digital theodolite, an Electronic Distance Measurement (EDM) device, and a microprocessor. By aiming an infrared or laser beam at a prism reflector, it simultaneously measures slope distances along with highly precise horizontal and vertical angles. The built-in computer instantly processes this field data to automatically calculate horizontal distances and exact 3D coordinates (Easting, Northing, and Elevation). A crucial feature of this instrument is its integrated **data logging** capability; it automatically records and stores all measurements and computed coordinates internally. This eliminates manual transcription errors and allows the digital data to be seamlessly transferred directly to CAD software for efficient, error-free map generation.



- **GPS:** The Global Positioning System (GPS) is a satellite-based navigation system used to determine the exact absolute coordinates (Latitude, Longitude, and Elevation) of points on the Earth's surface. It achieves this through **trilateration**, calculating the precise distances from a receiver to multiple

orbiting satellites to pinpoint a location. Because standard GPS is subject to atmospheric interference, civil engineering surveys utilize **Differential GPS (DGPS)**, which employs a stationary base receiver to broadcast real-time corrections to a roving unit, ensuring centimeter-level accuracy. Its **role in traversing** is essential for true geographical alignment. While a Total Station generates relative, "floating" coordinates based on an assumed local origin, integrating DGPS at the starting and closing stations of an open traverse "anchors" the entire survey into a global coordinate system, ensuring the final map overlays perfectly onto real-world geographical grids.

Procedure:

1. **Station Establishment:** Select the starting station (Station A) and subsequent stations (B, C, D...) for the open traverse route. Drive pegs to mark these points.
2. **GPS Initialization:** Place the GPS receiver exactly over Station A. Allow it to track satellites and record the precise coordinates (Easting, Northing, and Elevation).
3. **Total Station Setup:** * Set up the Total Station over Station A. Perform exact centering using the optical/laser plummet and level it using the electronic bubble.
 - Input the GPS-derived coordinates for Station A into the Total Station to establish the origin.
4. **Orientation:** To orient the instrument, you must define the North direction or backsight to a known point. If using only one GPS point at the start, orient the instrument to Magnetic North using an inbuilt or external compass and set the horizontal angle to 0°00'00".
5. **Traversing:**
 - Hold the prism pole perfectly vertical over the forward station (Station B).
 - Sight the prism with the Total Station and trigger a measurement.
 - The Total Station will automatically record the horizontal distance, horizontal angle, and vertical elevation, calculating the coordinates of Station B.
6. **Advancing the Traverse:** * Shift the Total Station to Station B and set it up.
 - Backsight to Station A to orient the instrument.
 - Move the prism to Station C, sight it, and record the measurements to determine Station C's coordinates.
 - Repeat this process until the final station of the route is reached.
7. **GPS Verification:** At the final station (or at random intermediate stations), use the GPS receiver again to capture the coordinates. Compare these GPS coordinates with the coordinates calculated by the Total Station to check the cumulative linear and angular accuracy of the open traverse.

Precautions :

1. Ensure the tripod legs are firmly pressed into the ground to prevent the instrument from settling during measurements.
2. The prism pole must be held perfectly vertical; always check the circular bubble on the pole before signaling the instrument operator to take a reading.
3. When using the GPS, wait until the receiver has locked onto a sufficient number of satellites to ensure a low Dilution of Precision (DOP) before saving the coordinate.
4. Protect the Total Station lens from direct sunlight and rain using an umbrella if necessary.

Observation Table:

Inst. Station	Target Station	Horiz. Angle	Horiz. Dist (m)	Vert. Dist (m)	TS Calc. Easting (X)	TS Calc. Northing (Y)	GPS Easting (X)	GPS Northing (Y)	Error / Remarks
A	B	00° 00' 00"	55.4	+ 1.20	5000	5000	5000.002	4999.998	Origin Point
B	C	145° 20' 10"	82.15	-0.45	5055.4	5000	-	-	Intermediate
C	D	210° 15' 30"	64.3	+ 0.85	5123.65	4954.12	-	-	Intermediate
D	E	175° 40' 00"	90.25	-1.1	5160.8	4890.3	5160.815	4890.28	Checkpoint

Conclusion: The open traverse route was successfully plotted using the Total Station, and the computed station coordinates closely matched the validation coordinates recorded by the GPS receiver.

*****Abbreviations-**

- **Inst. Station:** Instrument Station (The exact point where the Total Station is set up)
- **Horiz. Angle:** Horizontal Angle (The angle measured on the horizontal plane)
- **Horiz. Dist:** Horizontal Distance (The flat, level distance between two points)
- **Vert. Dist:** Vertical Distance (The difference in height/elevation between two points)
- **TS:** Total Station
- **Calc:** Calculated / Computed (Values generated by the instrument itself)
- **GPS:** Global Positioning System
- **DGPS:** Differential Global Positioning System (A more accurate version of GPS mentioned in the apparatus)
- **EDM:** Electronic Distance Measurement (The laser technology inside a Total Station that measures distance)
- **X:** Easting (The coordinate value on the horizontal/East-West axis)
- **Y:** Northing (The coordinate value on the vertical/North-South axis)
- **Z:** Elevation (The height above mean sea level)

EXPERIMENT 8

Practical 7: Chain Surveying

Aim: To conduct a chain survey of the given area and prepare a plan using a chain and offsets.

Apparatus Required:

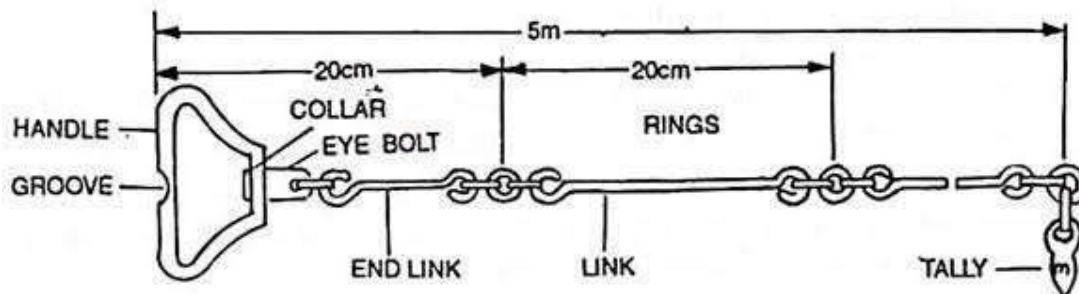
- Metric chain (30 m)
- Metallic or steel measuring tape (30 m)
- Ranging rods (min. 4 to 6)
- Arrows / Chain pins (typically 10)
- Wooden pegs and a hammer
- Cross-staff or Optical square (for setting right angles)
- Field book and pencil

Principle & Theory: The fundamental principle of chain surveying is **Triangulation**. This involves dividing the entire survey area into a network of well-conditioned triangles (triangles where no angle is less than 30° or greater than 120°). Since a triangle is the only simple polygon that can be completely plotted using only the lengths of its sides, chain surveying requires **only linear measurements**—no angular measurements are taken.

It is the simplest method of surveying and is best suited for small, fairly level, and open areas. To locate specific details (like buildings, trees, or road edges), **offsets** are measured. An offset is a lateral distance measured from the main chain line to the object. These can be perpendicular (90°) using a cross-staff, or oblique.

To ensure the accuracy of the framework, two specific types of lines are used:

- **Tie Lines:** Join subsidiary stations on the main lines to gather details that are too far from the main chain line.
- **Check Lines (Proof Lines):** Run across the triangles to verify the accuracy of the field measurements during plotting.



Procedure:

1. **Reconnaissance & Marking:** Inspect the area and securely drive wooden pegs into the ground to mark the starting and ending main stations (e.g., Station A and B).
2. **Ranging:** Place ranging rods along the line between the stations to ensure your measurement path is perfectly straight.
3. **Chaining:** Stretch the metric chain taut along the ranged baseline from the starting point to the end.
4. **Taking Offsets:** As you walk along the chain, use a cross-staff to establish a 90° -degree angle, and measure the perpendicular distance from the chain to nearby ground features using a metallic tape.
5. **Booking:** Immediately record the chainage (distance along the chain) and the corresponding offset distance (left or right) directly into the field book.

Precautions:

1. Unfold and fold the chain carefully to prevent links from getting tangled or bent.
2. Always pull the chain taut to minimize sag, keeping it as horizontal as possible on sloping ground.
3. Ensure ranging is perfectly straight before measuring long lines.
4. Record all measurements directly into the field book on the spot; never rely on memory.

Observation:

Length of main chain line: 30m

Length of check line:

Chain Section	Chainage (m)	Left Offset (m)	Right Offset (m)	Remarks	
Chain 1 (Line AB)	0	1.25	0.85	Start of Line AB / Station A	
	5	1.4	1.1	Wall	
	10	1.15	1.35	Wall	
	15	0.95	1.5	Wall (Check line connects here)	
	20	1.55	1.2	Wall	
	25	1.7	0.9	Wall	
	30	1.3	1.05	End of Line AB / Station B	
Chain 2 (Line BC)	0	1.3	1.05	Start of Line BC / Station B	
	5	1.1	1.45	Wall	
	10	0.85	1.75	Wall	
	15	1.45	1.9	Wall	
	20	1.6	1.55	Wall	
	25	1.85	1.25	Corner of brick wall (Left)	
	30	1.5	0.95	End of Line BC / Station C	
Chain 3 (Line CA)	0	1.5	0.95	Start of Line CA / Station C	
	5	1.75	1.15	Wall	
	10	1.9	1.4	Wall	
	15	1.55	1.65	Wall	
	20	1.2	1.8	Fence line starts (Right)	
	25	1.45	1.35	Wall	
	30	1.25	0.85	End of Line CA / Station A	

Conclusion: The chain survey of the specified area was successfully executed using the principles of triangulation, and an accurate plan of the area's features was plotted using the recorded offsets.